

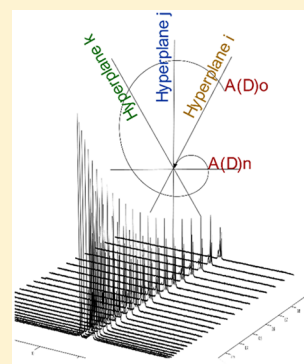
Algebraic Reconstruction Technique for Diffusion NMR Experiments. Application to the Molecular Weight Prediction of Polymers

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Supporting Information

ABSTRACT: Most of the algorithms employed in diffusion NMR are optimization methods based on diverse regularized methods such as Tikhonov's, which decomposes the multiexponential detected signal attenuation as a sum of mono exponential signals. Our approach uses projections over hyperplanes of the Hilbert space using a Laplace transform kernel, which is a special case of projection onto convex sets. This new application of an algebraic reconstruction technique for diffusion NMR experiments (dART) has been applied for the first time in both simulated and real systems, and then compared with established methods such as ITAMeD and TRAIIn. The new algorithm provides excellent results in systems with overlapped signals and more importantly performs more rapidly than any other one assayed. One of the main advantages is that the reported method does not need a regularization parameter, which allows one to explore the largest spaces. In addition, we have provided the calibration curve for weight-average M_w prediction of poly(propylene) polymers with no dependence on the solvent used.



1. INTRODUCTION

In the mid-1960s, Stejskal and Tanner used for the first-time NMR pulsed gradient spin echoes (PGSE) for the calculation of diffusion coefficients in isotropic media and from these estimated hydrodynamic radii.^{1–4} Diffusion NMR methodology holds a special position among the different experiments available to the NMR spectroscopist. While it provides invaluable information in molecular sizes^{5–8} and shapes,^{5,9} it can also be used to investigate polymers,^{4,10–14} organometallics,^{5,15,16} catalysis,^{17,18} nanoparticles,^{19–26} and host–guest systems,^{27–30} as well as to characterize complex mixtures.³¹ Other applications in biosystems and pharmaceuticals have been also provided.^{10,25,32–47}

The physical constant diffusion coefficient is estimated by the application of a PGSE technique, which is based on the signal attenuation ($E_{\text{diff}} = I/I_0$) during a corrected diffusion time Δ , which depends on the specific solution of the Stejskal–Tanner partial differential equation applied on a specific sequence, gradient shape and nucleus eq 1.⁴⁸

$$E_{\text{diff}} = e^{-D\gamma_{\text{eff}}^2 \delta^2 \sigma^2 g^2 \Delta} \quad (1)$$

For a continuous distribution of diffusion coefficients, $A(D)$, eq 1 could be replaced by the following integral eq 2 that describes the signal decay.

$$E_{\text{diff}}(g) = \int_0^\infty A(D) e^{-D\gamma_{\text{eff}}^2 \delta^2 \sigma^2 g^2 \Delta} dD \quad (2)$$

eq 2 could also be converted into discrete space through the inner product of the diffusion coefficient and the integral matrix of the multiexponential decay, which is named the Hilbert space eq 3.

$$E_{\text{diff}i} = \sum_{j=1}^n A_j e^{-D_j \gamma_{\text{eff}}^2 \delta^2 \sigma^2 g_i^2 \Delta} \quad (3)$$

where $E_{\text{diff}i} = E_{\text{diff}}(g_i)$, and $A_j = A(D_j)$. Subscripts i ($i = 1, 2, \dots, m$) and j ($j = 1, 2, \dots, n$) label PFG strengths and diffusion coefficient values, respectively. To simplify the mathematical treatment, eq 3 could be used as a linear algebraic form described by eq 4.

$$E_{\text{diff}} = \text{dHS} \cdot A \quad (4)$$

where E_{diff} is equal to $[E_{\text{diff}1}, E_{\text{diff}2}, \dots, E_{\text{diff}m}]^T$, the term A corresponds to $[A_1, A_2, \dots, A_n]^T$, and dHS is the $m \times n$ coefficient matrix corresponding to the diffusion Hilbert space defined by eq 5.

$$\text{dHS}_{ij} = e^{-D_j \gamma_{\text{eff}}^2 \delta^2 \sigma^2 g_i^2 \Delta} \quad (5)$$

We have recently introduced universal calibration curves (UCC) that allow the estimation of weight-average molecular weights in monodisperse polystyrene samples and globular folded proteins with no solvent dependence.^{47,49} In such samples the iterative thresholding algorithm for multiexponential decay (ITAMeD) method,⁵⁰ and conventional least mean square (LMS) routines were successfully applied. In the last two years, Zhang and co-workers have implemented the trust-region algorithm for the inversion method (TRAIIn),

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which is especially recommended for nonsymmetric distribution of diffusion coefficients.⁵¹

In general terms, the distribution of diffusion coefficients is estimated from experimental data, $E_{diff}(g)$, through the use of the inversion of the Laplace transform (ILT). This ILT is classified as an ill-posed problem which were originally introduced by Hadamard.⁵² Since the initial and boundary conditions are not well-defined, eventually strong vulnerability to noise and numerical instability have induced the emergence of different approaches. These can be divided in those based on total band shape, known as multivariate methods, and those run by single channel methods. Representatives for multivariate methods are DECRA,⁵³ MCR,⁵⁴ SCORE,⁵⁵ and OUTSCORE.⁵⁶ As single channel, we can regard Levenberg–Marquardt statistical method⁵⁷ and SPLMOD.⁵⁸ All these approaches consider the diffusion coefficient as monodisperse. Instead, other approaches that consider the diffusion coefficient as a distribution are CONTIN,⁵⁹ maximum entropy (MaxEnt)⁶⁰ and more recently PALMA.⁶¹ Importantly, most of them can be considered as Tikhonov regularization methods⁶² of the following least-squares minimization problem eq 6.

$$\|dHS \times A - E_{diff}\|_2^2 + \lambda f(A) \quad (6)$$

where λ is the sparsity promoted parameter and $f(A(D))$ is the function that penalized the overfit, and in overall constitutes the regularization term. The function $f(A(D))$ may impose the shape and width of the final solution, as is shown below.

Alternatively, instead of a regularization term, one can formulate the minimization problem as constrained, which is comparable to the TRAln method eq 7.

$$\begin{aligned} \min f(A) \\ \text{subject to: } \|dHS \times A - E_{diff}\| < \eta \end{aligned} \quad (7)$$

where η is always higher than zero, and it is based on an estimate of the experimental noise, which is related to the quality of the fitting.

Regarding the polydispersity index (PDI), which is one of the main variables in polymers science, it has previously been estimated through diffusion NMR experiments using either differential diffusion profiles⁶³ or applying gamma or log-normal distribution models.⁶⁴ In 2016, Urbanczyk et al. approached the problem of polydispersity by the use of a tailored regularization term,⁶⁵ and later on Delsuc and co-workers did the same by the application of the PALMA algorithm, which combines maximum sparsity and maximum entropy.⁶¹

The algebraic reconstruction technique (ART) is a class of iterative algorithm that can be considered as an iterative solver of a system of linear equations.⁶⁶ This algorithm was discovered in the 1930s,⁶⁷ and nowadays is extensively used in the field of image reconstruction.⁶⁸ The ART does not depend on any regularization parameter in contrast to the above-mentioned techniques, and it is very efficient in terms of computational time.⁶⁹ If we consider the diffusion NMR problem as a linear system eq 4, then, dHS is a full rank $m \times n$ matrix with $m \leq n$ and $E_{diff} \in R^m$. This method orthogonally projects on each step the last iterate onto the solution hyperplane of $dHS_i A(D) = E_{diff_i}$ and take this as the next iterate. Thus, the algorithm takes the form given in eq 8.

$$A_{k+1} = A_k + \frac{E_{diff_i} - \langle dHS_i, A_k \rangle}{\|dHS_i\|_2^2} \times dHS_i \quad (8)$$

In order to maximize the projection onto the hyperplane j , the distance between the real solution and the value taken from the hyperplane in the iteration k is evaluated through the eq 9. Thus, for the next iteration, the j th hyperplane with the largest distance is selected, and therefore the overfit in the calculation is prevented.

$$\begin{aligned} j(k) = \operatorname{argmax} \left(\frac{\operatorname{abs}(E_{diff_i} - dHS_i \times A_k)}{\|dHS_i\|} \right), \\ i = 1, 2, \dots, m; k = 1, 2, \dots; \end{aligned} \quad (9)$$

To improve the rate performance of the proposed application of ART, Nesterov's accelerated method was implemented.^{70,71} The ART algorithm itself has a rate of convergence proportional to $1/k$, where k is the number of iterations. When Nesterov's module is introduced, the rate of convergence is increased up to $1/k^2$ within the same number of iterations. For this reason, the following sequence was executed. Note that $\gamma_k \leq 0$.

$$\lambda_0 = 0, \lambda_k = \frac{1 + \sqrt{1 + 4\lambda_{k-1}^2}}{2}, \text{ and } \gamma_k = \frac{1 - \lambda_k}{\lambda_{k+1}} \quad (10)$$

Consequently, the ART was restructured following eq 11 for its application in diffusion NMR (dART).

$$A'_{k+1} = (1 - \gamma_k)A_{k+1} + \gamma_k A_k \quad (11)$$

It is important to mention, that the acceleration component performs a simple step of the maximum projection onto the hyperplane j from A_{k+1} to A'_{k+1} , and then it moves a little bit further than A_{k+1} in the direction given by the previous A_k .

With the dART algorithm already presented, we envisaged its application in diffusion NMR and specifically, for the quantitative determination of the diffusion coefficients and therefore, accurate prediction of molecular weights. The method has been tested on simulated and real samples such as a blend of monodisperse poly(propylene glycol) polymers in the validity range of Flory's law (i.e., absence of obstruction or concentration effects on diffusion measurements).^{72,73} We also present the comparison of the method with commonly applied algorithms such as ITAMeD and TRAln. It is worth stating that we always referred M_w in terms of averages since synthetic polymers have polydispersity.

2. EXPERIMENTAL SECTION

Simulations. Several simulated data sets, chosen to represent various analytical situations, were used for the evaluation of the algorithm. Set A consists in three monodisperse components with diffusion coefficients 0.02127×10^{-9} (D_1), 0.0638×10^{-9} (D_2), and $0.2127 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ (D_3), with respective intensities 1, 3, and 2. Set B is based on a binary mixture of 1:1 ratio. One of the simulated components was fixed at $0.0213 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, and the other one varied from 0.028×10^{-9} to $0.0521 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ in 12 steps. Set C is a wide distribution, simulated as a log-normal distribution centered at $0.2123 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, and presenting a PDI estimated to 2.30. Set D involves two monodisperse components with diffusion coefficients of 0.1611×10^{-9}

(D_1) and $0.3971 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ (D_2), with respective intensities 5:1.

Processing. The dART algorithm was implemented with the programming language of MATLAB (see [Codes S1–S3](#)). All computations were performed on a windows 64bit personal computer (PC) with an Intel i7-3770k @ 3.5 GHz and 24 GB of memory.

Samples. Poly(propylene glycol) polymers PPG450, PPG1000, PPG2000, PPG3200, and PPG5000 were purchased from the American Polymer Standards Corporation (Ohio, USA). Their corresponding M_w , M_n and PDI values are given in [Table S1](#). Benzene- d_6 was purchased from Eurisotop (Saint-Aubin, France), dried over CaH_2 , and vacuum transferred onto 3 Å molecular sieves prior to use. All other reagents and solvents were of commercial quality and were used without further purification. The UCC ([Figure 5](#) and [Table S3](#)) was built by measuring five PPG samples (PPG450, PPG1000, PPG2000, PPG3200, and PPG5000) that were prepared by dissolving 0.6 mg of each of the polymers in 0.6 mL of benzene- d_6 and then transferring those solutions into oven-dried 5 mm NMR tubes. The NMR sample constituted by two PPG polymers was prepared by just adding 0.6 mg of each polymer together with 0.6 mL of benzene- d_6 in an oven-dried 5 mm J. Young NMR tube.

NMR Spectroscopy. All measurements were performed as previously described.⁷⁴ The Δ and δ values varied from 75 to 100 ms and from 2.2 to 5 ms, respectively. The gradient strength was incremented in steps of 4%, so that 23 points could be used for regression analysis. The recovery delay was always set to 10 s. The number of scans per increment were 64 and typical experimental times were around 4 h. All experiments were run without spinning. To check reproducibility and lack of convection, three different measurements with different Δ were always carried out.⁷⁵ The contribution of convection to the calculated D values seems to be negligible since it remains always constant under the three different diffusion times assayed. ITAMeD solutions were obtained by the use of the algorithm provided by Urbanczyk et al.⁶⁵ The number of iterations was always set to 10000 and the sparsity-promoting λ to 10^{-5} . TRAIIn solutions were obtained using the algorithm provided by Xu and Zhang.⁵¹

3. RESULTS AND DISCUSSION

To test the vulnerability to noise of our dART method, we run Monte Carlo simulations on set A. Overall, 30 simulations were performed with random Gaussian noise applied of 0.0, 0.1, 0.5 and 1.0%. [Figure 1](#) presents the results obtained on the simulated experiment A consisting in the superposition of three monodisperse species, separated by less than a factor of 10 in diffusion coefficients. [Figures S2 and S3](#) show the results obtained for ITAMeD and TRAIIn alternatives.

The time elapsed in these simulations of 4096 points in the D space, were 2 min for dART and 14 and 21 for ITAMeD and TRAIIn, respectively. Overfit was avoided in the case of ITAMeD and TRAIIn by the introduction of a sparsity promoted value equal to $1e^{-5}$, and an α parameter equal to 1.02, respectively. On the contrary, dART does not need the interplay of such parameters. The fact that ITAMeD uses the L1-norm regularization gives rise to sharper solutions.

All algorithms presented the same accuracy in the reconstruction of D values at the lowest noise level of 0.0 and 0.1%. For levels above 0.5%, the errors obtained were above 5.6% which are comparable to previous results.⁶⁵ [Table](#)

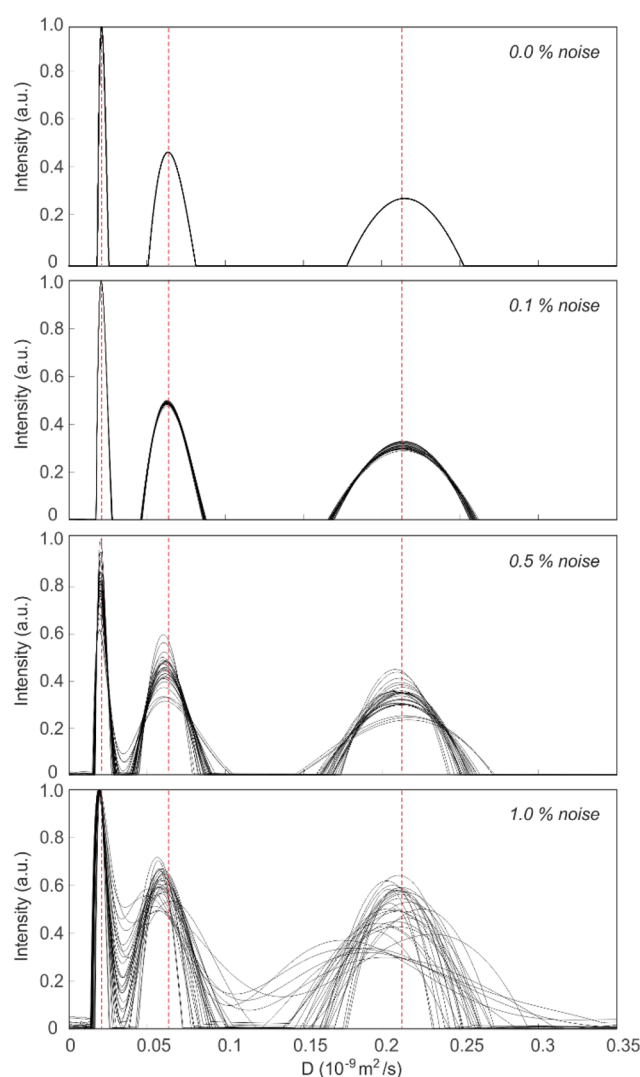


Figure 1. dART processing of the simulated data set with added Gaussian noise levels of 0.0, 0.1, 0.5, and 1.0%. Reference D -values are marked with dotted lines.

[1](#) presents synthetic results that highlights in bold those algorithms that afford lower errors compared to those originally defined in the set of data. Extensive results are presented in the Supporting Information (see [Figures S2 and](#)

Table 1. Quality of Reconstruction of Signals of Set A at Diffusion Coefficients D_1 , D_2 and D_3 , with Different Algorithms for Various Noise Levels (in %)^a

	noise level	dART	ITAMeD	TRAIIn
D_1	0.1	2.56	2.25	7.13
	0.5	5.62	2.25	7.13
	1.0	9.16	2.25	7.30
D_2	0.1	1.72	3.70	6.14
	0.5	4.17	3.81	6.30
	1.0	8.22	5.44	7.83
D_3	0.1	0.35	2.04	0.56
	0.5	0.27	2.02	0.13
	1.0	2.92	4.77	0.12

^aQuality is computed as $(|D_{\text{sim}} - D_{\text{exp}}|/D_{\text{exp}})$ expressed in %. For each noise level, the highest quality results are outlined in bold.

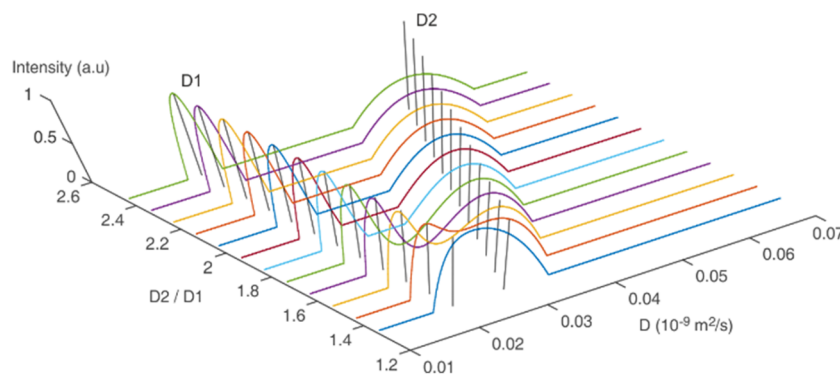


Figure 2. Results of dART processing on set B which is based on two peaks at varying ratio of diffusion coefficients. The vertical lines represent reference values set in the simulation.

S3 and Table S2). Interestingly, the solutions for the first peak (large molecule or small D -value) is very narrow whereas the solutions for the second and third peaks (small to medium size molecules) become very broad. This is due to the fact that the first peak contains signal in the whole I/I_0 attenuation, whereas the other two only have signal in the first part of the attenuation and therefore the algorithms have more difficulties in locate the solution.

The strength of our algorithm was tested with three more approaches based on simulation. On one hand, we applied the dART method on the differentiation of a binary mixture of 1:1 ratio (set B). One of the simulated components was fixed at $0.0213 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, and the other one varied from 0.028×10^{-9} to $0.0521 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ in 12 steps. Each simulation was performed with additional Gaussian noise on the level of 0.1%. Remarkably, the dART method could solve both components with relative errors below $\Delta D_{\text{rel}} = 0.001 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, when the separation of D values was above $0.0095 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, that is in the second step (Figure 2). TRAI_n behaved similarly than the dART algorithm (Figure S4), whereas ITAMeD (Figure S5) showed less component resolution than both of them.

The second approach (set C) was based on the reconstruction of a broad Gaussian line simulating a polydisperse polymer with a PDI of about 2. Figure 3 shows

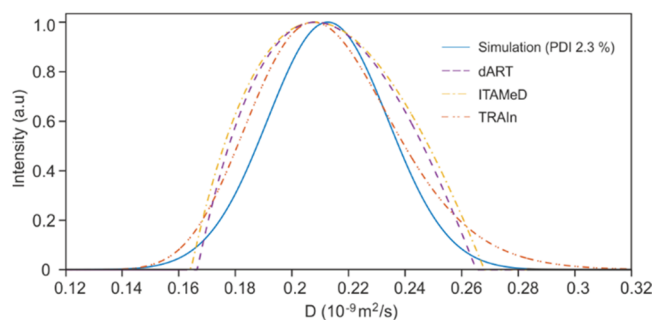


Figure 3. dART Reconstruction of a polydisperse Gaussian profile. For comparison issues, ITAMeD and TRAI_n results are also shown.

the reconstruction of this signal profile with the three methods. TRAI_n performs reasonably well, being the method of choice for polydisperse distributions. On the contrary, dART and ITAMeD lead to a deficient reconstruction of the line-shape with a poor prediction of its width. Nevertheless, the three methods predict exactly the same D coefficient ($0.2081 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$) with quality values below 2.0%.

As it has been proven, dART is not sensitive to broadness, and although minor distortions are observed, do not affect the final reconstruction which is an important issue when solving polydisperse systems.

In order to prove that the methods analyzed herein, and in particular dART, retrieved the right ratio in their solutions, the data set D involving two monodisperse components in a 5:1 ratio was studied. At the three levels of Gaussian noise of 0.1, 0.5 and 1.0%, the obtained ratio were in all the cases 5.1:1, 4.5:1 and 4.2:1, respectively, what corroborates that when the attenuation I/I_0 contains signal belonging to the two components, i.e. when the components are of similar size or do have similar D -values, the simulated ratio in where they exist fits excellently with the real one. It is important to mention, that in contrast to set A, where the D -values were separated by a factor of 10, in the data set D this difference was reduced down to 2. Figures 4, S7, and S8 show the comparison of the methods at the three level of noise of 0.5, 0.1 and 1.0%, respectively. We are currently stretching the scope of our method in these terms and in mixtures based on more than two components. The results from these investigations will be reported in due course.

It is known that experimental data are quite different from simulated one where usually coexist sharp and large diffusion distributions, sprinkle with instrumental artifacts and nonstationary noise. To test the behavior of dART on real samples, the technique was first applied on PGSE experiments measured on five poly(propylene glycol) (PPG) polymers in benzene- d_6 with referenced molecular weight (from PPG450 to PPG 5000) and referenced polydispersity (from 1.07 to 1.16). Detailed information on the PPG-samples studied, which includes their M_w , M_n , and PDI values, is given in Table S1. Figure 5 presents the $D\eta$ values of the five polymers samples with respect to their commercial molecular masses in a log–log plot.

The D values employed are in all the cases averages that arise from the monitoring of various resonances (Table S3). The straight line is the result of fitting eq 12 to this data, with slopes of -0.6071 ($R^2 = 0.9989$), -0.5894 ($R^2 = 0.9976$), -0.5953 ($R^2 = 0.9982$) and -0.5970 ($R^2 = 0.9985$) for dART, LMS, ITAMeD and TRAI_n, respectively. The variables c , f_s , N_A , and K correspond to size factor, shape factor, Avogadro's number, and a proportionality nondefined constant that relates molecular weight with viscosity.⁴⁹ Importantly, the spreading around the theoretical curve is in all the cases very weak, with R^2 values between 0.998 and 0.999.

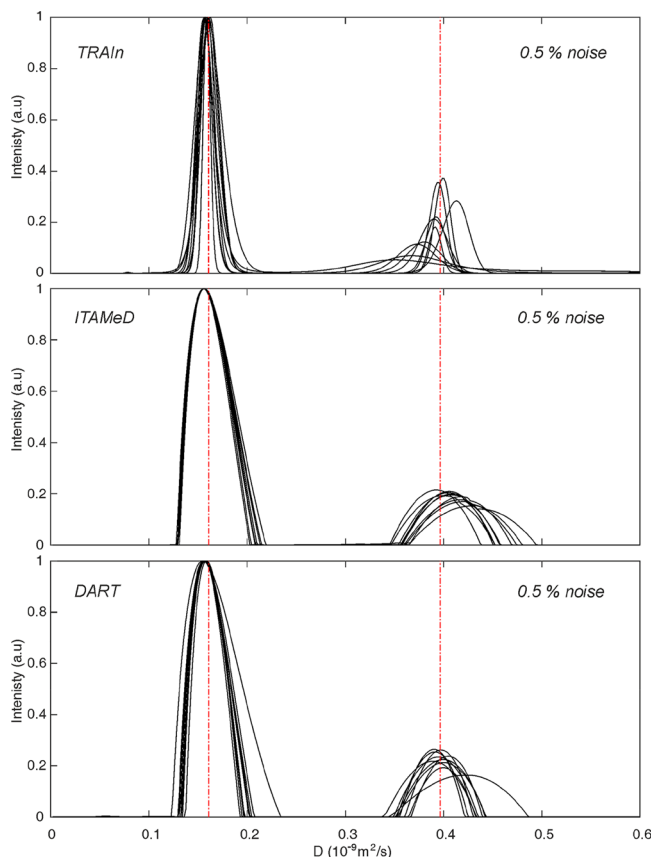


Figure 4. dART, ITAMeD, and TRAIn processing on set D. Thirty diffusion coefficients distribution were calculated with reference D-values marked with dotted lines. The simulated attenuations were performed with 0.5% of noise level.

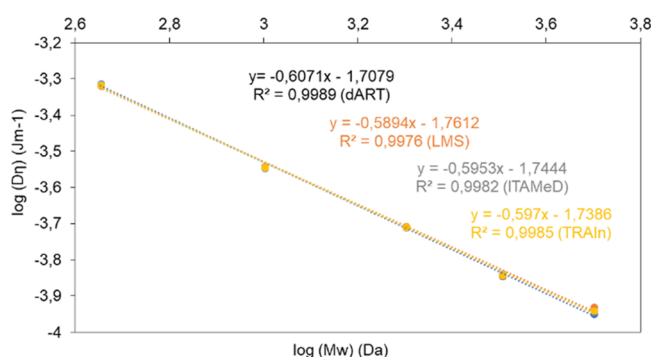


Figure 5. Calibration curve for M_w prediction in PPG polymers accessed through the use of diverse algorithms.

$$\log(D\eta) = -\frac{1}{3\beta} \log(M_w) - \frac{1}{\beta} \log \left(\frac{c_f^s K^{1-\beta} \eta^{\beta(\beta-1)} \left(\frac{3}{4\rho N_A \pi} \right)^{1/3}}{k_B T} \right) \quad (12)$$

The term η^β , and thus β , describes structural requirements of the solvent in the presence of certain functional groups of the solute. This liquid structuring and its associated entropy are generally responsible for much of the behavior of macromolecular systems.^{47,49} The combination of the slope of these

curves with eq 12 yields, as mentioned above, a power law behavior with a β -value of 0.549 (dART), which corresponds to an average fractal dimension d_F of 1.647 ($1/\alpha$) in Delsuc nomenclature.

This new relation represents the first calibration curve for weight-average M_w prediction of poly(propylene glycol) polymers with no dependence on the solvent used. A similar curve, in this case for polystyrene polymers has been already described, in which the β value found of 0.586 suggested that both type of systems induce a similar structuring pattern of the solvent molecules over these polymer chains.

In addition, to estimate the feasibility of dART, results of PDIs retrieved by the three methods, i.e. dART, ITAMeD, and TRAIn, are illustrated in Tables S5–S7, showing similar results in all the analyzed PPG samples with difference errors between 1.8 and 12.1%.

The performance of our dART method was additionally analyzed by using a binary blend of monodisperse PPG polymers in benzene- d_6 . The results afforded by the others regularization methods such as ITAMeD and TRAIn are also shown for comparison issues and proves the strength of our approach regarding the accurate calculation on D-values, and eventually the correct prediction of the corresponding molecular weights. We found that the PPG mixture was well-suited for the tests of multiexponential fitting methods, because all the protons in the two PPG systems are no-dependent on molecular weight and resonate at the same frequency. Thus, the only way of distinguishing between them using NMR is to perform PGSE experiments and solve their diffusion coefficients. Analysis of the overall intensity attenuation showed that dART, ITAMeD, and TRAIn work excellently well (Table 2), though dART recovered both D values with the lowest errors among them, below 1.3%.

Table 2. Diffusion Values ($10^{-9} \text{ m}^2 \text{ s}^{-1}$) at Room Temperature (294 K) and Diff Errors (%) Calculated through the Different Methods

	dART	ITAMeD	TRAIn
PPG450	0.7566	0.7124	0.7157
diff (%) ^a	0.4	5.4	4.9
PPG5000	0.1806	0.1817	0.1808
diff (%) ^a	1.3	2.4	1.8

^aWith respect to D-values of 0.7475 and $0.1831 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ for PPG450 and PPG5000, respectively.

Assuming the error bars on the “true” diffusion coefficients are of the order of 10%, there is no real difference between methods. The two D-values together with the already developed calibration curve for PPG samples (Figure 5), were employed for the weight-average M_w predictions. To our delight, the estimated M_w compared to real values fitted excellently well with errors below 8.0% (Table 3). On the contrary, when the predictions are intended with alternative algorithms, less accurate weights are obtained (Table S4), supporting dART as the most reliable for this specific poly(propylene glycol) blend.

Importantly, the time of convergence for the proposed method was evaluated. For this purpose, the data set corresponding to PPG3200 was considered. The algorithm dART allowed the fastest convergence within 0.12 s in 20 iterations, whereas ITAMeD for example, needed five seconds

Table 3. dART Diffusion Values ($10^{-9} \text{ m}^2 \text{ s}^{-1}$)^a and UCC-Predicted Weight-Average M_w

PS	D	$D\eta$ (10^{-3}) ^b	$Av-M_w$ (Da) ^c	diff (%) ^d
450	0.7566	0.1275	437	2.8
5000	0.1806	0.0058	4380	7.3

^a¹H PFG-STE measurements were performed at room temperature (294 K). ^bThe viscosity value employed was η $0.646 \times 10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$. ^cEstimated via the dART calibration curve $y = -0.6071x - 1.7079$ ($R^1 = 0.9989$). ^dConsidering commercial values of 450 and 5000 Da.

to reach 1000 iterations with a sparsity promoted parameter of 1×10^{-5} (Figure S6).

Finally, the concentration of the sample was also evaluated in order to understand if increasing the concentration of the PPG sample would alter the performance of dART. We measure the medium size polymer PPG2000 at three increasing concentrations, i.e., 0.6, 6, and 60 mg in 0.6 mL of benzene-*d*₆, that is at 0.1, 1.0 and 10.0 wt %, respectively. As expected, the algorithm performed excellently regardless of the concentration of the sample. About the results retrieved, the diffusion coefficients at the two lowest concentrations were almost the same with values of 0.3052 and $0.3043 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, respectively, suggesting Flory's infinite dilution behavior in both cases. At 10.0 wt %, the obtained *D*-value was $0.2259 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, which arises from both an increased viscosity and an aggregation effect. To calculate molecular weights at 1 and 10 wt % we need to know the viscosities of both solutions. To circumvent this problem, we monitor the intensity attenuations of benzene in both samples. Because the r_H value for benzene can be determined ($r_H = 1.53 \text{ \AA}$), the measured *D* values obtained together with the Stokes–Einstein equation, afford realistic estimates of the benzene PPG-containing viscosities of 0.6462 and $0.7369 \times 10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$, respectively. Within these values the predicted M_w were calculated with the help of the calibration curve (Figure 5), obtaining values of 1948 and 2635 Da, which clearly indicates that at higher concentration the prediction significantly deviates with respect to the theoretical value of 2000 Da.

The PDI values retrieved by dART in the three samples of 0.1, 1.0, and 10.0 wt % were 1.030, 1.048, and 1.049, respectively, which strongly support that the algorithm is not affected by any concentration effect.

4. CONCLUSIONS

Diffusion NMR still in development and it is probably on complex mixtures or polymer blends where stretches all his power. In fact, a simple fit of the data to the basic evolution equations usually fails in providing a faithful analysis of the data or solving the number of components and their molecular masses. For these reasons, one must use the inverse Laplace transform for the analysis. In this work, we have introduced for the first time an algebraic reconstruction technique (dART) in diffusion NMR to solve the inverse problem. The proposed method does not need sparsity promote parameter neither alpha value and therefore allows to explore largest spaces. This new algorithm has been compared with established methods such as ITAMeD or TRAln on both simulated and real systems, providing excellent results and with the lowest times of convergence. An additional advantage is its ability to work well with large diffusion spaces. On the contrary, dART does not provide the best results with highly polydisperse polymers

and provide broader solutions that for instance ITAMeD that uses the regularization L1-norm. In addition, we have provided a calibration curve for weight-average M_w prediction of polypropylene polymers with no dependence on the solvent used, in which the β value of 0.549 has been established. The model is presented in a desktop application available at <http://www2.ual.es/NMRMBC/downloads/UCCPolymerPrediction.rar>.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpca.8b08584.

Experimental section, complete NMR diffusion data, programming codes, and algorithm solutions (PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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■ REFERENCES

- (1) Stejskal, E. O.; Tanner, J. E. Spin diffusion measurements: Spin echoes in the presence of a time dependent field gradient. *J. Chem. Phys.* **1965**, *42*, 288–292.
- (2) Tanner, J. E. Use of the stimulated echo in NMR diffusion studies. *J. Chem. Phys.* **1970**, *52*, 2523–2526.
- (3) Chen, A.; Wu, D.; Johnson, C. S. Determination of molecular weight distributions for polymers by diffusion-ordered NMR. *J. Am. Chem. Soc.* **1995**, *117*, 7965–7970.
- (4) Johnson, C. S. Diffusion ordered nuclear magnetic resonance spectroscopy: principles and applications. *Prog. Nucl. Magn. Reson. Spectrosc.* **1999**, *34*, 203–256.
- (5) Macchioni, A.; Ciancaleoni, G.; Zuccaccia, C.; Zuccaccia, D. Determining accurate molecular sizes in solution through NMR diffusion spectroscopy. *Chem. Soc. Rev.* **2008**, *37*, 479–489.
- (6) Li, D.; Kagan, G.; Hopson, R.; Williard, P. G. Formula weight prediction by internal reference diffusion-ordered NMR spectroscopy (DOSY). *J. Am. Chem. Soc.* **2009**, *131*, 5627–5634.
- (7) Neufeld, R.; Stalke, D. Accurate molecular weight determination of small molecules via DOSY-NMR by using external calibration curves with normalized diffusion coefficients. *Chem. Sci.* **2015**, *6*, 3354–3364.
- (8) Evans, R.; Deng, Z.; Rogerson, A. K.; McLachlan, A. S.; Richards, J. J.; Nilsson, M.; Morris, G. A. Quantitative interpretation of diffusion-ordered NMR spectra: can we rationalize small molecule diffusion coefficients? *Angew. Chem., Int. Ed.* **2013**, *52*, 3199–3202.
- (9) Evans, R.; Dal Poggetto, G.; Nilsson, M.; Morris, G. A. Improving the interpretation of small molecule diffusion coefficients. *Anal. Chem.* **2018**, *90*, 3987–3994.

- (10) Viel, S.; Capitani, D.; Mannina, L.; Segre, A. Diffusion-ordered NMR spectroscopy: A versatile tool for the molecular weight determination of uncharged polysaccharides. *Biomacromolecules* **2003**, *4*, 1843–1847.
- (11) Barrere, C.; Mazarin, M.; Giordanengo, R.; Phan, T.; Thevand, A.; Viel, S.; Charles, L. Molecular weight determination of block copolymers by pulsed gradient spin echo NMR. *Anal. Chem.* **2009**, *81*, 8054–8060.
- (12) Li, W.; Chung, H.; Daeffler, C.; Johnson, J. A.; Grubbs, R. H. Application of ^1H DOSY for facile measurement of polymer molecular weights. *Macromolecules* **2012**, *45*, 9595–9603.
- (13) Kuz'mina, N. E.; Moiseev, S. V.; Krylov, V. I.; Yashkir, V. A.; Merkulov, V. A. Quantitative determination of the average molecular weights of dextrans by diffusion ordered NMR spectroscopy. *J. Anal. Chem.* **2014**, *69*, 953–959.
- (14) Lewinski, P.; Sosnowski, S.; Kazmierski, S.; Penczek, S. L-lactide polymerization studied by ^1H NMR with diffusion ordered spectroscopy (DOSY). "One NMR tube experiment" providing: Conversion, polymer structure, M_n and M_w . *Polym. Chem.* **2015**, *6*, 4353–4357.
- (15) Pregosin, P. S.; Kumar, P. G. A.; Fernández, I. Pulsed gradient spin-echo (PGSE) diffusion and ^1H , ^{19}F heteronuclear overhauser spectroscopy (HOESY) NMR methods in inorganic and organometallic chemistry: Something old and something new. *Chem. Rev.* **2005**, *105*, 2977–2998.
- (16) Avram, L.; Cohen, Y. Diffusion NMR of molecular cages and capsules. *Chem. Soc. Rev.* **2015**, *44*, 586–602.
- (17) Schober, K.; Hartmann, E.; Zhang, H.; Gschwind, R. M. ^1H DOSY spectra of ligands for highly enantioselective reactions – a fast and simple NMR method to optimize catalytic reaction conditions. *Angew. Chem., Int. Ed.* **2010**, *49*, 2794–2797.
- (18) Horeglad, P.; Litwinska, A.; Zukowska, G. Z.; Kubicki, D.; Szczepaniak, G.; Dranka, M.; Zachara, J. The influence of organosuperbases on the structure and activity of dialkylgallium alkoxides in the polymerization of rac-lactide: the road to stereo diblock PLA copolymers. *Appl. Organomet. Chem.* **2013**, *27*, 328–336.
- (19) Kohlmann, O.; Steinmetz, W. E.; Mao, X. A.; Wuelfing, W. P.; Templeton, A. C.; Murray, R. W.; Johnson, C. S. NMR diffusion, relaxation, and spectroscopic studies of water soluble, monolayer-protected gold nanoclusters. *J. Phys. Chem. B* **2001**, *105*, 8801–8809.
- (20) Van Lokeren, L.; Maheut, G.; Ribot, F.; Escax, V.; Verbruggen, I.; Sanchez, C.; Martins, J. C.; Biesemans, M.; Willem, R. Characterization of titanium dioxide nanoparticles dispersed in organic ligand solutions by using a diffusion-ordered spectroscopy-based strategy. *Chem. - Eur. J.* **2007**, *13*, 6957–6966.
- (21) Van Lokeren, L.; Cartuyvels, E.; Absillis, G.; Willem, R.; Parac-Vogt, T. N. Phosphoesterase activity of polyoxomolybdates: diffusion ordered NMR spectroscopy as a tool for obtaining insights into the reactivity of polyoxometalate clusters. *Chem. Commun.* **2008**, 2774–2776.
- (22) Marega, R.; Aroulmoji, V.; Dinon, F.; Vaccari, L.; Giordani, S.; Bianco, A.; Murano, E.; Prato, M. Diffusion-ordered NMR spectroscopy in the structural characterization of functionalized carbon nanotubes. *J. Am. Chem. Soc.* **2009**, *131*, 9086–9093.
- (23) Marega, R.; Aroulmoji, V.; Bergamin, M.; Feruglio, L.; Dinon, F.; Bianco, A.; Murano, E.; Prato, M. Two-dimensional diffusion-ordered NMR spectroscopy as a tool for monitoring functionalized carbon nanotube purification and composition. *ACS Nano* **2010**, *4*, 2051–2058.
- (24) Canzi, G.; Mrse, A. A.; Kubiak, C. P. Diffusion-ordered NMR spectroscopy as a reliable alternative to TEM for determining the size of gold nanoparticles in organic solutions. *J. Phys. Chem. C* **2011**, *115*, 7972–7978.
- (25) de Kort, D. W.; van Duynhoven, J. P. M.; Hoebe, F. J. M.; Janssen, H. M.; Van As, H. NMR nanoparticle diffusometry in hydrogels: enhancing sensitivity and selectivity. *Anal. Chem.* **2014**, *86*, 9229–9235.
- (26) Thielemann, D. T.; Wagner, A. T.; Lan, Y.; Oña-Burgos, P.; Fernández, I.; Rösch, E.; Kölmel, D. K.; Powell, A. K.; Bräse, S.; Roesky, P. W. Peptoid ligated pentadecanuclear yttrium and dysprosium hydroxy clusters. *Chem. - Eur. J.* **2015**, *21*, 2813–2820.
- (27) Cohen, Y.; Avram, L.; Frish, L. Diffusions-NMR-spektroskopie in der supramolekularen und kombinatorischen chemie: Ein alter parameter – neue erkenntnisse. *Angew. Chem.* **2005**, *117*, 524–560.
- (28) Cohen, Y.; Avram, L.; Frish, L. Diffusion NMR spectroscopy in supramolecular and combinatorial chemistry: An old parameter – new insights. *Angew. Chem., Int. Ed.* **2005**, *44*, 520–554.
- (29) Pastor, A.; Martinez-Viviente, E. NMR spectroscopy in coordination supramolecular chemistry: A unique and powerful methodology. *Coord. Chem. Rev.* **2008**, *252*, 2314–2345.
- (30) Han, S.; Ma, Z.; Hopson, R.; Wei, Y.; Budil, D.; Gulla, S.; Moulton, B. Postsynthetic modification of a coordination compound with a paddlewheel motif via click reaction: DOSY and ESR studies. *Inorg. Chem. Commun.* **2012**, *15*, 78–83.
- (31) Hamdoun, G.; Guduff, L.; van Heijenoort, C.; Bour, C.; Gandon, V.; Dumez, J.-N. Spatially encoded diffusion-ordered NMR spectroscopy of reaction mixtures in organic solvents. *Analyst* **2018**, *143*, 3458–3464.
- (32) Crutchfield, C. A.; Harris, D. J. Molecular mass estimation by PFG NMR spectroscopy. *J. Magn. Reson.* **2007**, *185*, 179–182.
- (33) Wang, C. K.; Northfield, S. E.; Swedberg, J. E.; Harvey, P. J.; Mathiowetz, A. M.; Price, D. A.; Liras, S.; Craik, D. J. Translational diffusion of cyclic peptides measured using pulsed-field gradient NMR. *J. Phys. Chem. B* **2014**, *118*, 11129–11136.
- (34) Dutta, A. R.; Sekar, P.; Dvoyashkin, M.; Bowers, C. R.; Ziegler, K. J.; Vasenkov, S. Single-file diffusion of gas mixtures in nanochannels of the dipeptide l-alal-val: High-field diffusion NMR study. *J. Phys. Chem. C* **2016**, *120*, 9914–9919.
- (35) Pagès, G.; Bonny, A.; Gilard, V.; Malet-Martino, M. Pulsed field gradient NMR with sigmoid shape gradient sampling to produce more detailed diffusion ordered spectroscopy maps of real complex mixtures: Examples with medicine analysis. *Anal. Chem.* **2016**, *88*, 3304–3309.
- (36) Trefi, S.; Routaboul, C.; Hamieh, S.; Gilard, V.; Malet-Martino, M.; Martino, R. Analysis of illegally manufactured formulations of tadalafil (Cialis) by ^1H NMR, 2D DOSY ^1H NMR and Raman spectroscopy. *J. Pharm. Biomed. Anal.* **2008**, *47*, 103–113.
- (37) Balayssac, S.; Gilard, V.; Delsuc, M. A.; Malet-Martino, M. DOSY NMR, a new tool for fake drug analyses. *Spectrosc. Eur.* **2009**, *21*, 10–14.
- (38) Vayse, J.; Balayssac, S.; Gilard, V.; Desoubdizanne, D.; Malet-Martino, M.; Martino, R. Analysis of adulterated herbal medicines and dietary supplements marketed for weight loss by DOSY ^1H -NMR. *Food Addit. Contam., Part A* **2010**, *27*, 903–916.
- (39) Balayssac, S.; Retailleau, E.; Bertrand, G.; Escot, M.-P.; Martino, R.; Malet-Martino, M.; Gilard, V. Characterization of heroin samples by ^1H NMR and 2D DOSY ^1H NMR. *Forensic Sci. Int.* **2014**, *234*, 29–38.
- (40) Smejkalova, D.; Piccolo, L. Host-guest interactions between 2,4-dichlorophenol and humic substances as evaluated by ^1H NMR relaxation and diffusion ordered spectroscopy. *Environ. Sci. Technol.* **2008**, *42*, 8440–8445.
- (41) Li, D.; Keresztes, I.; Hopson, R.; Williard, P. G. Characterization of reactive intermediates by multinuclear diffusion-ordered NMR spectroscopy (DOSY). *Acc. Chem. Res.* **2009**, *42*, 270–280.
- (42) Lamm, J. H.; Niermeier, P.; Mix, A.; Chmiel, J.; Neumann, B.; Stamm, H. G.; Mitzel, N. W. Mechanism of host-guest complex formation and identification of intermediates through NMR titration and diffusion NMR spectroscopy. *Angew. Chem., Int. Ed.* **2014**, *53*, 7938–7942.
- (43) Cao, R.; Nonaka, A.; Komura, F.; Matsui, T. Application of diffusion ordered- ^1H -nuclear magnetic resonance spectroscopy to quantify sucrose in beverages. *Food Chem.* **2015**, *171*, 8–12.
- (44) Ge, W.; Zhang, J. H.; Pedersen, C. M.; Zhao, T.; Yue, F.; Chen, C.; Wang, P.; Wang, Y.; Qiao, Y. DOSY NMR: A versatile analytical chromatographic tool for lignocellulosic biomass conversion. *ACS Sustainable Chem. Eng.* **2016**, *4*, 1193–1200.

- (45) Wilkins, D. K.; Grimshaw, S. B.; Receveur, V.; Dobson, C. M.; Jones, J. A.; Smith, L. Hydrodynamic radii of native and denatured proteins measured by pulse field gradient NMR Techniques. *Biochemistry* **1999**, *38*, 16424–16431.
- (46) Jones, J. A.; Wilkins, D. K.; Smith, L. J.; Dobson, C. M. Characterisation of protein unfolding by NMR diffusion measurements. *J. Biomol. NMR* **1997**, *10*, 199–203.
- (47) Arrabal-Campos, F. M.; Aguilera-Sáez, L. M.; Fernández, I. A diffusion NMR method for the prediction of the weight-average molecular weight of globular proteins in aqueous media of different viscosities. *Anal. Methods* **2019**, *11*, 142–147.
- (48) Sinnaeve, D. The Stejskal–Tanner equation generalized for any gradient shape – An overview of most pulse sequences measuring free diffusion. *Concepts Magn. Reson., Part A* **2012**, *40A*, 39–65.
- (49) Arrabal-Campos, F. M.; Oña-Burgos, P.; Fernández, I. Molecular weight prediction with no dependence on solvent viscosity. A quantitative pulse field gradient diffusion NMR approach. *Polym. Chem.* **2016**, *7*, 4326–4329.
- (50) Urbanczyk, M.; Bernin, D.; Kozminski, W.; Kazimierczuk, K. Iterative thresholding algorithm for multiexponential decay applied to PGSE NMR data. *Anal. Chem.* **2013**, *85*, 1828–1833.
- (51) Xu, K.; Zhang, S. Trust-region algorithm for the inversion of molecular diffusion NMR data. *Anal. Chem.* **2014**, *86*, 592–599.
- (52) Hadamard, J. *Lectures on the Cauchy Problem in Linear Partial Differential Equations*; Yale University Press: New Haven, CT, 1923.
- (53) Antalek, B. Using pulsed gradient spin echo NMR for chemical mixture analysis: How to obtain optimum results. *Concepts Magn. Reson.* **2002**, *14*, 225–258.
- (54) Van Gorkom, L. C. M.; Hancewicz, T. Analysis of DOSY and GPC-NMR experiments on polymers by multivariate curve resolution. *J. Magn. Reson.* **1998**, *130*, 125–130.
- (55) Nilsson, M.; Morris, G. A. Speedy component resolution: An improved tool for processing diffusion-ordered spectroscopy data. *Anal. Chem.* **2008**, *80*, 3777–3782.
- (56) Colbourne, A. A.; Meier, S.; Morris, G. A.; Nilsson, M. Unmixing the NMR spectra of similar species – vive la difference. *Chem. Commun.* **2013**, *49*, 10510–10512.
- (57) Press, W. H.; Teukolsky, S.; Vetterling, W.; Flannery, B. P. *Numerical recipes: The art of scientific computing*; Cambridge University Press: New York, 1992.
- (58) Provencher, S. W.; Vogel, R. H. Regularization Techniques for Inverse Problems in Molecular Biology. *Numerical treatment of inverse problems in differential and integral equations* **1983**, 304–319.
- (59) Provencher, S. W. CONTIN: A general purpose constrained regularization program for inverting noisy linear algebraic and integral equations. *Comput. Phys. Commun.* **1982**, *27*, 229–242.
- (60) Delsuc, M. A.; Malliavin, T. E. Maximum entropy processing of DOSY NMR spectra. *Anal. Chem.* **1998**, *70*, 2146–2148.
- (61) Cherni, A.; Chouzenoux, E.; Delsuc, M. A. PALMA, an improved algorithm for DOSY signal processing. *Analyst* **2017**, *142*, 772–779.
- (62) Tikhonov, A. N. Solution of incorrectly formulated problems and the regularization method. *Soviet Math. Dokl.* **1963**, *4*, 1035–1038.
- (63) Vieville, J.; Tanty, M.; Delsuc, M. A. Polydispersity index of polymers revealed by DOSY NMR. *J. Magn. Reson.* **2011**, *212*, 169–173.
- (64) Roding, M.; Bernin, D.; Jonasson, J.; Sarkka, A.; Topgaard, D.; Rudemo, M.; Nyden, M. The gamma distribution model for pulsed-field gradient NMR studies of molecular-weight distributions of polymers. *J. Magn. Reson.* **2012**, *222*, 105–111.
- (65) Urbanczyk, M.; Bernin, D.; Czuron, A.; Kazimierczuk, K. Monitoring polydispersity by NMR diffusometry with tailored norm regularisation and moving-frame processing. *Analyst* **2016**, *141*, 1745–1752.
- (66) Gordon, R.; Bender, R.; Herman, G. T. Algebraic reconstruction techniques (ART) for three-dimensional electron microscopy and x-ray photography. *J. Theor. Biol.* **1970**, *29*, 471–481.
- (67) Kaczmarz, S. Angenäherte auflösung von systemen linearer gleichungen. *Sciences Mathématiques* **1937**, *35*, 355–357.
- (68) Herman, G. T. *Fundamentals of computerized tomography. image reconstruction from projections*; Springer: New York, 2009.
- (69) Zhdanov, A. I. The method of augmented regularized normal equations. *Comp. Math. and Math. Phys.* **2012**, *52*, 194–197.
- (70) Nesterov, Y. E. A method for solving the convex programming problem with convergence rate $O(1/k^2)$. *Soviet Mathematics Doklady* **1983**, *27*, 372–376.
- (71) Nesterov, Y. E. Accelerating the cubic regularization of Newton's method on convex problems. *Math. Program.* **2007**, *112*, 159–181.
- (72) Flory, P. J. *Principles of polymer chemistry*; Cornell University Press: New York, 1953.
- (73) Edwards, S. F.; Doi, M. *The theory of polymer dynamics*; Oxford University Press: New York, 1986.
- (74) Arrabal-Campos, F. M.; Alvarez, J. D.; Garcia-Sancho, A.; Fernández, I. Molecular weight prediction in polystyrene blends. Unprecedented use of a genetic algorithm in pulse field gradient spin echo (PGSE) NMR. *Soft Matter* **2017**, *13*, 6620–6626.
- (75) Barbosa, T. M.; Rittner, R.; Tormena, C. F.; Morris, G. A.; Nilsson, M. Convection in liquid-state NMR: expect the unexpected. *RSC Adv.* **2016**, *6*, 95173–95176.